

KITEWRIGHT LIFT · DESIGN DECISION RECORD

Kitewright Lift TRL 2 review decisions

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DOCUMENT CONTROL

Document ID	KWL-DDR-001	Version	0.1
Status	Draft	Date	2026-10-03
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/kitewright-lift		
Source	docs/decisions/0001-trl2-review-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-03	Amish Chadha	TRL 2 review items decided under Amish's pre-approvals of 2026-10-03

0001: TRL 2 review decisions

- **Date:** 2026-10-03
- **Status:** accepted

■ Context

The TRL 2 review (docs/REVIEW.md , TRL 2 section) and the open questions in KWL-PRB-001 raised the items below. On 2026-10-03 Amish wrote: "start with the first 14 repos from the list of 29 projects. I pre-approve the batch runs along with any recommendations you come up with. I also accept any cost overruns or variations from the assumed scope cost." He then wrote: "Proceed with the remaining 15 scaffolds", under the same pre-approval, which covers Kitewright Lift. Every design recommendation below is therefore decided as recommended. Choices that touch safety take the conservative option, with the evidence that would relax them stated. Partners and regions are the first candidates to approach, not agreements. Requirements that are not met or at risk on paper (R1 and R2 on mass, R4 on hover time) are not decided here; they are posed to Amish in docs/REVIEW.md under "Decisions for Amish".

Kitewright Core, Lift and Range are designed in parallel. Where Lift depends on a sibling (the Core stack and payload mount, the ColdCell packs), the sizes used here are interface assumptions; they are listed in docs/REVIEW.md under "Cross-repo actions" for the sibling designs to confirm.

■ Options considered

Table 1 lists the options for each item and the one chosen.

■ Decision

Table 1. Items decided on 2026-10-03 under Amish's pre-approvals.

#	Item	Options	Decision	What would relax a safety choice
1	Rotor layout	(a) flat hexacopter; (b) flat octocopter; (c) coaxial X8 (four arms, a motor above and below each tip)	(c), conservative: losing any one motor leaves its partner on the same arm, so the aircraft can still hover (thrust to weight with one motor out 2.20 at sea level and 1.51 at 5,000 m on paper). Four arms fold into a smaller bundle than six or eight. Costs about 4 % more hover power than a flat layout of the same span	Not relaxed: motor-out tolerance is a requirement (R5)
2	Arm fold	(a) arms fold up, held by a lock pin; (b) arms fold sideways; (c) arms fold down, lift pressing each arm up against a fixed stop	(c), conservative: in flight the thrust holds the arm against a stop bridge machined into the hinge block, so a lock pin left out cannot let an arm fold under power. The ball-lock pin only stops the arm drooping on the ground or in a hard landing, and carries a red flag so a missing pin is visible	Not relaxed. A proof load on CalRig of each hinge at 1.5 times the full-thrust moment is required before first flight (item 13)
3	Propeller and motor class	28, 30 or 32 inch propellers; integrated or separate controllers	30 inch folding carbon propellers on 1,240 mm motor spacing (115 mm tip gap); about 100 KV motors of the 10 kg static thrust class; separate 80 A controllers strapped to the arms, so motor leads never cross the hinge	Not a safety choice
4	Power bus and packs	ColdCell as designed (LiFePO ₄ or sodium-ion); a lithium-ion pack	Two ColdCell LiFePO ₄ packs, 16S2P of 26650 power cells (51.2 V, about 307 Wh each), used unchanged as the problem statement's constraint requires. The hover time this gives is posed to Amish as decision 2 in docs/REVIEW.md , not decided here	Charging and cold limits stay as ColdCell sets them
5	Tether power	Low-voltage tether; fixed high-voltage tether with onboard conversion; adaptive tether voltage	Fixed 400 V DC from an isolated ground supply over a 60 m tether of two 0.75 mm ² conductors; a 4 kW isolated DC-DC converter on a payload plate; a pull-release breakaway at the aircraft. Conservative: isolated supply with an insulation monitor, residual-current protection and an emergency stop at the ground; tether hover limited to 50 m. Stays clear of adaptive tether voltage (Elistair US11059580B2)	A ground-fault and touch-current test of the complete system could allow tether hover above 50 m; the isolation and monitoring stay
6	Winch	Design a winch now; reserve space only	No winch is designed or published until its own patent screen is done. The payload bay and rails take a winch module later; nothing in this design depends on one	Not a safety choice
7	Line-and-float release	Hook release; servo pin release; drop the line attached to the aircraft	A servo pin release that fails closed holds a webbing sling round a foam float; the float's line is tied to the float only and is never attached to the aircraft. Conservative: drops from at least 10 m above the person, so downwash (about 7.5 m/s estimated at 10 m) does not push them under	Downwash measured over water at lower heights could lower the 10 m drop height
8	Payload interface	Lift-specific mount; the Core payload mount	The Kitewright Core payload mount under the bottom plate: two slotted rails, a ball-lock payload pin through both rails and the payload's lug, and the DS-014 socket at the back. Interface sizes are assumptions (Cross-repo actions)	Not relaxed: the pin must be in and flagged before arming

#	Item	Options	Decision	What would relax a safety choice
9	Landing gear	Folding gear; fixed skids; detachable legs	Fixed skid gear, 340 mm skids 450 mm apart, hub underside 500 mm up. Fixed gear stays on for transport (the folded aircraft still fits the R9 case) and the folded motors clear the skids by 17 mm	Not a safety choice
10	Keep-out and hover heights	Set by test later; set now	Set now, conservatively: nobody within 15 m of the aircraft when armed except over a person in water for a drop; drops from 10 m or higher; no hover over people on snow slopes or rooftops	Downwash and noise measurements in the TRL 4 trials
11	First co-design candidate	A state disaster force; a volunteer mountain rescue team; a hazard research group	The Himalayan Rescue Association, Nepal, as the first candidate to approach (not agreed): it runs aid posts on the trekking routes and works with helicopter rescue at altitude	Not a safety choice
12	First high-altitude test region	Ladakh; Sikkim; Nepal's Khumbu	Sea-level trials first, then the Khumbu region of Nepal with the co-design candidate, as the first candidate (not agreed); flights only with the civil aviation authority's permission	Not a safety choice
13	Proof before flight	Fly and see; bench proof first	AltiRig as the first candidate rig for motor and propeller thrust at 5,000 m density (R3); CalRig as the first candidate rig for proof loads: each arm hinge at 1.5 times the full-thrust moment (117 N m) and the tether breakaway pull	Not relaxed
14	Prototype budget	Raise the budget; keep it	<code>budget_usd</code> kept at USD 5,000 as a value-engineering target; the estimated cost of USD 5,739 is reported against it	Not a safety choice

■ Consequences

- KWL-PRC-001, KWL-REQ-001 and KWL-PRB-001 are updated to version 0.2 with these decisions.
- The constructable design (KWL-DDR-002) and the sizing (KWL-CAL-001) follow items 1 to 9.
- R4 (hover time) is not met with the LiFePO4 packs of item 4 and R1 is exceeded by 0.5 kg with a 5 kg payload; both are posed to Amish, with options, in `docs/REVIEW.md`.
- Items 5, 7, 8, 10 and 13 appear as safety stops in the build plan (KWL-BLD-001).